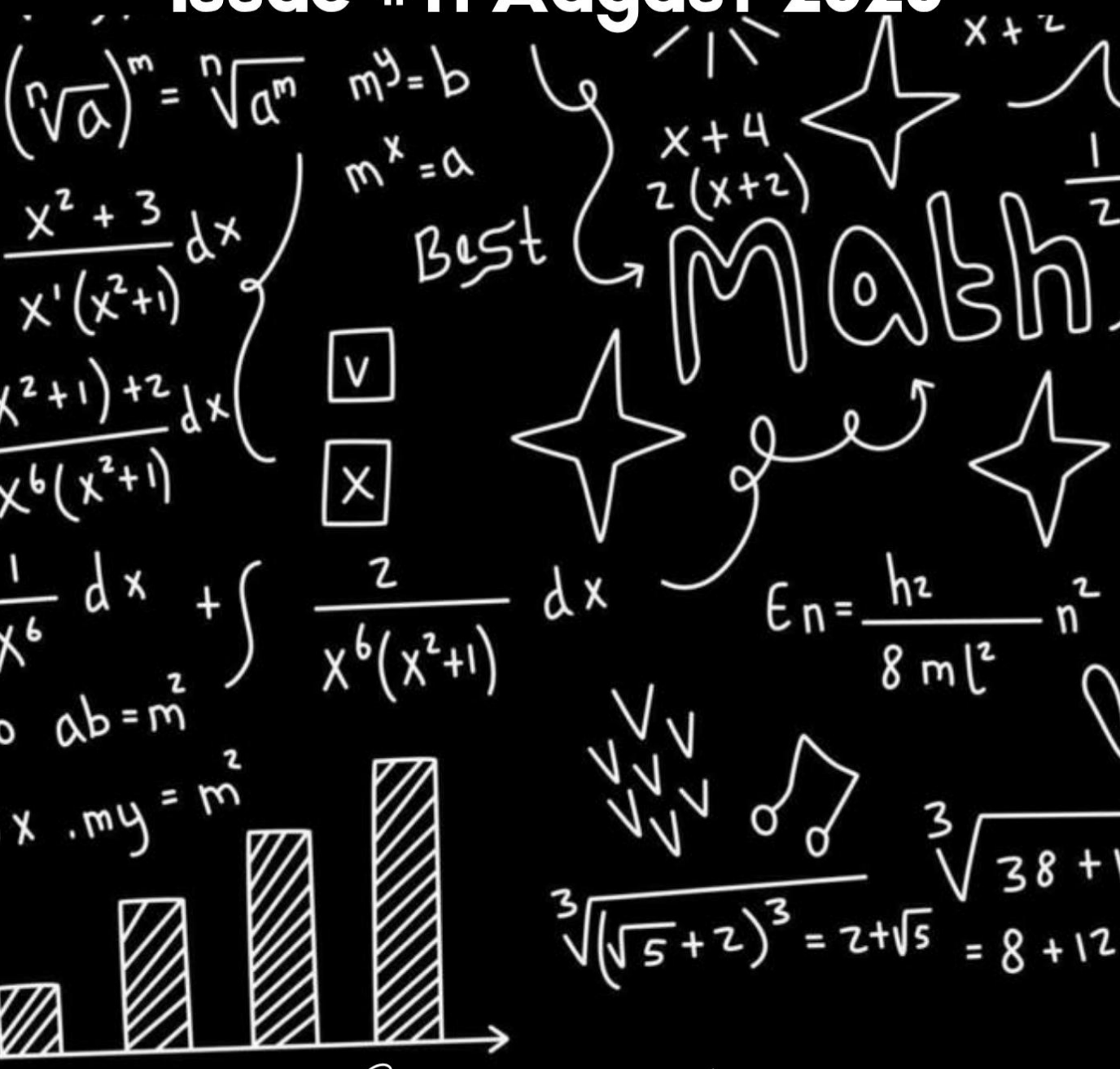
The Youth Science Journal Of Iraq

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Satvik M. Bhure

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A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

Question I: How can a simple equation calculate the age of the universe?

– Taim Muhammed, 16

The age of the universe can be determined using a fundamental formula from classical physics: time equals distance divided by speed (t= d/s, which can be written as t= $1/H_o$). By measuring the rate at which distant galaxies are receding from Earth, we can begin to calculate!

To visualize this concept, consider tracking a standard journey. If the current distance of a vehicle and its constant speed are both known, calculating its departure time is straightforward. When astronomer Edwin Hubble conducted observations in 1929, he discovered that distant galaxies are moving away from Earth. Furthermore, he noted a direct proportion: galaxies located twice as far away travel at twice the speed. Because velocity is distance divided by time, the equation can be rearranged to isolate time. When a galaxy's specific distance and expansion velocity are entered into this formula, the distance units cancel out completely. This mathematical simplification leaves a singular measurement of time.

However, a raw calculation using only this baseline formula assumes that the universe has expanded at an entirely constant rate throughout history. In reality, the velocity of this expansion has fluctuated across different epochs due to the competing influence of two massive cosmic forces: gravity and dark energy.

During the first billion years after the Big Bang, the collective gravitational pull of all matter in the universe acted as a deceleration mechanism, slowing the expansion. Eventually, a poorly understood force known as dark energy became dominant, acting as an accelerator that sped up the expansion. Because the universe transitioned from slowing down to speeding up, scientists must apply a correction factor to the primary equation.

To achieve an exact pinpoint age, astrophysicists combined this core equation with precise satellite data collected from the cosmic microwave background. Adjusting the formula to account for these historical variations in speed yields a highly accurate result. This refined calculation ultimately proves that the universe is approximately 13.8 billion years old.

Question II: Why do computers depend on math that was invented long before computers?

– *Roza Abdullah, 13*

Computers are one of the most important inventions of the modern world. They can perform millions of calculations in a very short amount of time and are used in almost every aspect of life. However, many people may wonder why computers depend so much on mathematics that was developed long before computers even existed. The simple answer is that mathematics provides the rules and logic that allow computers to work.

Mathematics has been studied for thousands of years. Long before computers, people developed ideas about numbers, patterns, logic, geometry and calculations. These ideas later became useful when scientists began creating machines that could process information. For example, binary numbers, which are based on only 0 and 1, are an important part of how modern computers store and process information. Although binary mathematics existed before modern computers, it became extremely useful for electronic machines.

Computers also depend on logic and algorithms. An algorithm is basically a set of steps used to solve a problem. This idea is not limited to computers. Humans have been using mathematical methods to solve problems for a very long time. When computers were developed, these methods could be turned into instructions that a machine could follow.

Another important reason is that mathematics is based on rules that do not change easily. A mathematical equation developed hundreds of years ago still follows the same principles today. This makes mathematics a reliable foundation for computer science.

Therefore, computers did not create the need for mathematics. Instead, they took old mathematical ideas and used them in a new and much more powerful way. Mathematics existed first, and computers became a tool for applying it much faster than humans could.

Question III: What do we know about prime numbers?

– Azad Daryan, 13

2, 3, 5, 7, 11… At some point or another, we've all seen this sequence of numbers. Many of us are familiar with it. We know it as the sequence of *prime numbers.* But what *are* they?

Easy. They're numbers that have two factors: one and themselves. The smallest and only even prime number is 2; the rest are all odd. As we go up, prime numbers get rarer. But it doesn't stop there.

Prime numbers are useful. If a number isn't prime, it can be written as a product of primes. This notation can be used to determine whether a number is another number raised to a power, or to find HCFs and LCMs. We can consider them building blocks, in some way. What's so special about them?

We have some information about prime numbers, but they're a mystery we haven't yet solved. They tie into several different hypotheses and conjectures that have remained unsolved for decades.

Picture this: You keep counting and checking the factors of all numbers you reach. You want to find primes. So, you keep counting upwards, checking your odd numbers. Eventually, it gets boring. The primes are rare. It seems almost *impossible* to find them. You give up. It only seems rational that, after a certain point, prime numbers stop existing, right?

Wrong. Sometime around 300 BC, famous mathematician Euclid proved that there are an infinite number of prime numbers as we scale up to infinity. His method to prove this goes as follows:

Let's assume there's a finite number of prime numbers: $p_1, p_2, p_3, \ldots, p_n$. Multiply all these prime numbers together to get a new number *N*, then add 1. *N*, like all integers greater than 1, must either be prime or not. If you think it is, that makes our life easier. If you think it isn't, then let's show you that it is.

If you divide this prime number by any other prime number (which are like building blocks), you will get a remainder of one because of the 1 we added at the end. This means that there should be a number multiplied by 1 to get *N*, making N a prime number because it has no other factors.

Another famous mystery regarding prime numbers is Goldbach's Conjecture. Every even number above two can be written as the sum of two primes. You can have a try - go on, do your best! Prove it wrong.

4 = 2 + 2, 6 = 3 + 3, 12 = 7 + 5

And so forth. This shows us that we still have a lot to learn about these numbers, and a lot to study. However, we could have a lot of fun doing so.

Question IV: What is the history of zero?

– Rami Karzan, 14

The concept of a number meaning nothing was unknown during most of history. The Greeks, Romans, and Egyptians had number systems; however, they all started from 1.

During 300 BC, the Babylonians used a placeholder for the number 0. However, this was not a number but instead something to distinguish numbers such as 14 and 104. The Maya civilization used a shell-shaped glyph representing 0 in their calendars during the 4th century CE.

Between the 5th and 7th century CE, Indians were the first to truly use 0 as a number. In 628 CE, the mathematician Brahmagupta became the first person to establish a definition of 0, stating, "zero is the result of subtracting a number from itself."

During the Islamic Golden Age, many Islamic mathematicians, such as Al-Khwarizmi, further developed 0 and integrated it into the Hindu-Arabic number system. In the 12th century, it spread to Europe through trade. Moreover, the Italian mathematician Fibonacci popularized the Hindu-Arabic number system.

Nowadays, the number 0 is important for everything, from simple algebra to calculus, and without the number 0, we would not have the vast advancements in mathematics and technology that we have today.

The simplest use of zero is to distinguish numbers such as 10, 100, and 10,000. It also separates the negative and positive numbers and is used to solve algebraic equations.

More complex uses of 0 include calculus, as it is essential to limits, derivatives, and integration. Moreover, it makes binary possible so that computers can operate.

And although zero is immensely important in mathematics, it also made a contribution to philosophy; the void of 'nothingness' that was once impossible to understand could now be envisioned using zero.

So zero is everywhere – in thought, experience, and especially in math. Considering the hands-on experience of math, is there anything that does not require zero, whether subtly or blatantly?

Question V: Why is the distance formula so important?

– *Maryam Ahmed, 15*

The distance formula mentioned here is: $\sqrt{[(x2-x1)^2+(y2-y1)^2]}$

Based on the famous Pythagorean theorem, the distance formula is used to calculate the distance between two points on a plane with their x (horizontal) and y (vertical) coordinates or the alternative measurement. It was published in 1731 by French mathematician Alexis Claraut who was the first person to write the formula as we now know it, his formula builds of the work of ancient mathematicians Euclid and Pythagoras.

Widely used every day all around us, it drives technologies like navigation systems and many machine learning algorithms each built for a different task. It works behind the scenes to yield distance measurements required for more complex calculations that are later used for many things. Although the usage of the distance formula extends to hundreds of different fields, a very familiar usage of the distance formula is in video games!

Video games are now becoming the most popular form of entertainment in the world, and many games have a world that players can explore which, naturally, requires the player to be able to move. This seemingly abstract system relies on the distance formula! Let's look at a video game world, whether three dimensional or two dimensional. Players can move at a certain speed and move a certain distance while markers show the remaining distance to a point of interest. But how do the developers behind it do this? Well they use the distance formula! Places and players in a world are treated as two coordinate points on a plane, and by using this formula the distance can be calculated.

This shows how deeply math is integrated into our world. One simple formula derived centuries ago is now one of the most important to modern math and its integration into technology.

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And we thank you, reader, for supporting our
mission to help the Iraqi youth where there
has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

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